兴泰盈科

XT T1000 Android API Document

Version	Modified by	Date	Description
V1.0.0	Kelly	2023-05-15	First edition
V2.0.0	Kelly	2023-08-04	Simplify API
V2.1.0	Kelly	2023-08-18	Add based on Bitmap display
V2.2.0	Kelly	2023-08-29	Support for local update
V2.3.0	Kelly	2023-09-06	Support for 13.3-inch screen
V2.3.1	Kelly	2023-09-12	Fix the 42 full-screen mirror problem
V2.3.2	Kelly	2023-09-27	Fix the 13.3 full-screen mirror problem
V2.4.0	Kelly	2023-10-08	1.31.2 monochrome 8951 2.Fix only one valid issue when brushing drawings for multiple USB devices with identical vid and pid 3.Add the return device information
V2.4.1	Kelly	2023-10-27	Fix returned device information
V2.5.0	Kelly	2023-11-28	Support for 25.3-inch Kaleido
V2.6.0	Kelly	2023-12-11	Support for 13.3-inch Kaleido
V2.7.0	Kelly	2024-01-26	1.Add several interfaces (display image, firmware upgrade, waveform upgrade) 2.Support 28 monochrome 3.Perfect 31.2 monochrome 8951

V2.7.1	Kelly	2024-03-20	Improve 25.3 E6 adaptation (different T1000)
V2.8.0	Kelly	2024-04-12	1.Support 42 monochrome 8951 2.Improve the mirror image processing method
V2.9.0	Kelly	2024-04-29	1.Add several interfaces (temperature, VCom) 2.Display the image interface adjustment
V2.9.1	Kelly	2024-05-30	Optimize the monochrome screen series image display,support for dithering output display
V2.9.2	Kelly	2024-06-11	Repair the E5 display problem and correct the monochrome screen series grayscale problem
V2.10.0	Kelly	2024-06-26	Support for 25.3 E6V3 (latest firmware)
V2.10.1	Kelly	2024-09-19	Known problem optimization
V2.10.2	Kelly	2024-09-30	Known problem fix
V2.10.3	Kelly	2024-12-16	IT8951 Support to support the return of the waveform version
V2.10.4	Kelly	2025-01-16	Optimize data transmission
V2.10.5	Kelly	2025-06-06	Add Eink E6 dithering method
V3.0.0	Kelly	2025-07-10	Cumulative Updates and Detailed Improvements
V3.0.0	Kelly	2025-09-15	Document update: Added temperature range description for different Waveform modes

Catalog

1 Introduction

1.1 Purpose

This document describes the API library xt-t1000-3.0.0.jar and so library of Android and T1000 communication. Based on USB communication, realize image loading, image display and other functions.

1.2 Document range

✓xt-t1000-3.0.0.jar引入

✓API definition

✓API instruction

✓API input and output parameters

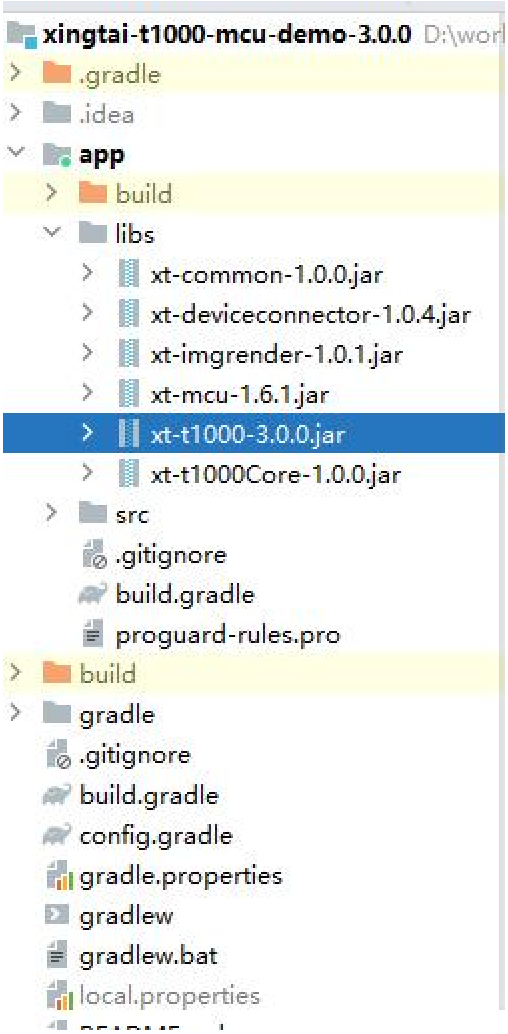
1.3 Expect readers

Development, testing, and related approvals

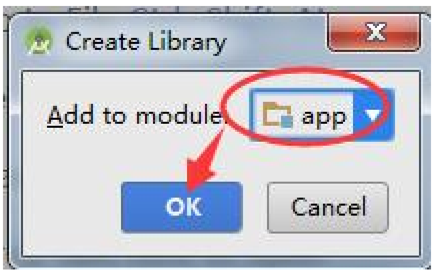
2 Android Studio import

1.Copy the provided xt-t1000-3.0.0.jar file to the app-> libs folder.

2.Right-click on the added jar file and select Add As Library。



3.Select the corresponding item module, and then click OK。



4.Copy demand copy so library to src/main/jniLibs。

3 Description of the API interface function

3.1 Open

Function: public int open()

Instruction: Open the usb connection

Return value type	Returned value specification	Examples

int	0 Success, other reference status code	

3.2 Close

Function： public void close()

Instruction： Close the usb connection

Return value type	Returned value specification	Examples
void	No	

3.3 Gets the connection status

Function： public int getState()

Instruction： Get the usb connection status

Return value type	Returned value specification	Examples
int	0 Success, other reference status code	

3.4 Set the screen type

Function： public void setType(int type)

Instruction： Initializes the screen type

Return value type	Returned value specification	Examples
int	type,screen type	0 - 25.3" EG 1 - 31.2" EC 2 - 25.3" EC 3 - 42" EC 4 - 25.3" E6(Deprecated) 5 - 28" EG 6 - 25.3" E5 7 - 13.3" EC 8951 8 - 25.3" EK 9 - 13.3" EK 10 - 28" EC 11 - 25.3" E6V2(Deprecated) 12 - 25.3" E6V3 20 - 31.2" EC 8951 21 - 42" EC 8951

3.5 Get device information

Function： public T1000DeviceInfo getT1000DeviceInfo()

Instruction： Get device information

Return value type	Returned value specification	Examples
T1000DeviceInfo	Not empty, indicating successfully obtaining device information, wide and high resolution of the returned device, supported display mode, firmware version, etc	

3.6 Get Temperature

Function： public int getTemperature()

Instruction： Obtain the sensor temperature

Return value type	Returned value specification	Examples
int	temperature value	

3.7 Set Temperature

Function： public int setTemperature(int temp)

Instruction： Set the temperature

Input parameter type	Introduction	Example
int	temp,temperature scale	25

Return value type	Returned value specification	Examples
int	0 Success, other reference status code	

3.8 Get VCom

Function： public int getVCom()

Instruction： Obtain the VCom

Return value type	Returned value specification	Examples
int	VCom value,unit： mV	1780mV representation -1.78v

3.9 Set VCom

Function： public int setVCom(int value)

Instruction： Set the VCom

Input parameter type	Introduction	Example
int	value,VCom value,unit： mV	1780mV representation -1.78v

Return value type	Returned value specification	Examples
int	0 Success, other reference status code	

3.10 Display(Deprecated)

Function： public int display(int type, String path,boolean enableDither, int startX, int startY, int imgW, int imgH, int waveform)

Instruction： Display image

Input parameter type	Introduction	Example

int	type,screen type	0 - 25.3" EG 1 - 31.2" EC 2 - 25.3" EC 3 - 42" EC 4 - 25.3" E6(Deprecated) 5 - 28" EG 6 - 25.3" E5 7 - 13.3" EC 8951 8 - 25.3" EK 9 - 13.3" EK 10 - 28" EC 11 - 25.3" E6V2(Deprecated) 12 - 25.3" E6V3 20 - 31.2" EC 8951 21 - 42" EC 8951
String	path,Image path	
boolean	enableDither,Whether to enable shake point, default false, color screen true.monochrome screen true display is better, but the performance is lower	
int	startX,Start the position in the picture X direction, and adjust the picture display position	0
int	startY,Start the position in the picture Y direction, and adjust the picture display position	0
int	width,The width of the image	Global brush, take the width of the screen that the width of the screen is equal to the width of the picture, local brush take the width of the picture
int	height,The image of the height	Global brush, take the height of the screen that the height of the screen is equal to the height of the picture, local brush take the height of the picture
int	waveform mode	Color screen 0, other 0,1,2

Return value type	Returned value specification	Examples
int	0 Success, other reference status code	

3.11 Display(Bitmap)(Deprecated)

Function： public int display(int type, Bitmap bitmap,boolean enableDither, int startX, int startY, int imgW, int imgH, int waveform)

Instruction： Display image

Input parameter	Introduction	Example

type		
int	type,screen type	0 - 25.3" EG 1 - 31.2" EC 2 - 25.3" EC 3 - 42" EC 4 - 25.3" E6(Deprecated) 5 - 28" EG 6 - 25.3" E5 7 - 13.3" EC 8951 8 - 25.3" EK 9 - 13.3" EK 10 - 28" EC 11 - 25.3" E6V2(Deprecated) 12 - 25.3" E6V3 20 - 31.2" EC 8951 21 - 42" EC 8951
Bitmap	bitmap,Bitmap	
boolean	enableDither,Whether to enable shake point, default false, color screen true.monochrome screen true display is better, but the performance is lower	
int	startX,Start the position in the picture X direction, and adjust the picture display position	0
int	startY,Start the position in the picture Y direction, and adjust the picture display position	0
int	width,The width of the image	Global brush, take the width of the screen that the width of the screen is equal to the width of the picture, local brush take the width of the picture
int	height,The image of the height	Global brush, take the height of the screen that the height of the screen is equal to the height of the picture, local brush take the height of the picture
int	waveform mode	Color screen 0, other 0,1,2

Return value type	Returned value specification	Examples
int	0 Success, other reference status code	

3.12 Clear Screen(Deprecated)

Function： public int clearDisplay(int type, int startX, int startY, int imgW, int imgH, int waveform)

Instruction： Clear the screen, as shown in white

Input parameter type	Introduction	Example

int	type,screen type	0 - 25.3" EG 1 - 31.2" EC 2 - 25.3" EC 3 - 42" EC 4 - 25.3" E6(Deprecated) 5 - 28" EG 6 - 25.3" E5 7 - 13.3" EC 8951 8 - 25.3" EK 9 - 13.3" EK 10 - 28" EC 11 - 25.3" E6V2(Deprecated) 12 - 25.3" E6V3 20 - 31.2" EC 8951 21 - 42" EC 8951
int	startX,Start the position in the picture X direction, and adjust the picture display position	0
int	startY,Start the position in the picture Y direction, and adjust the picture display position	0
int	width,The width of the image	Global brush, take the width of the screen that the width of the screen is equal to the width of the picture, local brush take the width of the picture
int	height,The image of the height	Global brush, take the height of the screen that the height of the screen is equal to the height of the picture, local brush take the height of the picture
int	waveform mode	Color screen 0, other 0,1,2

Return value type	Returned value specification	Examples
int	0 Success, other reference status code	

3.13 Load Image

Function： public int loadImage(LoadParams loadParams)

Instruction： Load image,after executing this function, you need to call 3.14 to display the image

Input parameter type	Introduction	Example
LoadParams	path,Image loading parameters	LoadParams loadParams = new LoadParams.Builder(bitmap, false) .setStartX(startX) .setStartY(startY) .setImgW(width) .setImgH(height)

		.setDisplayMode(waveformMode).build();

Return value type	Returned value specification	Examples
int	0 Success, other reference status code	

3.14 Load Image(String)

Function： public int loadImage(String path,boolean enableDither, int startX, int startY, int imgW, int imgH, int waveform)

Instruction： Load image,after executing this function, you need to call 3.12 to display the image

Input parameter type	Introduction	Example
String	path,Image path	
boolean	enableDither,Whether to enable shake point, default false, color screen true.monochrome screen true display is better, but the performance is lower	
int	startX,Start the position in the picture X direction, and adjust the picture display position	0
int	startY,Start the position in the picture Y direction, and adjust the picture display position	0
int	width,The width of the image	Global brush, take the width of the screen that the width of the screen is equal to the width of the picture, local brush take the width of the picture
int	height,The image of the height	Global brush, take the height of the screen that the height of the screen is equal to the height of the picture, local brush take the height of the picture
int	waveform mode	Color screen 0, other 0,1,2

Return value type	Returned value specification	Examples
int	0 Success, other reference status code	

3.15 Load Image(Bitmap)

Function： public int display(Bitmap bitmap,boolean enableDither, int startX, int startY, int imgW, int imgH, int waveform)

Instruction： Load image,after executing this function, you need to call 3.12 to display the image

Input parameter type	Introduction	Example
Bitmap	bitmap,Bitmap	
boolean	enableDither,Whether to enable shake point, default false, color screen true.monochrome screen true display is better, but the performance is lower	
int	startX,Start the position in the picture X direction, and adjust the picture display position	0
int	startY,Start the position in the picture Y direction, and adjust the picture display position	0
int	width,The width of the image	Global brush, take the width of the screen that the width of the screen is equal to the width of the picture, local brush take the width of the picture
int	height,The image of the height	Global brush, take the height of the screen that the height of the screen is equal to the height of the picture, local brush take the height of the picture
int	waveform mode	Color screen 0, other 0,1,2

Return value type	Returned value specification	Examples
int	0 Success, other reference status code	

3.16 Load Image(byte[])

Function： public int display(bytep[] pixels, int startX, int startY, int imgW, int imgH, int waveform)

Instruction： Load image,after executing this function, you need to call 3.12 to display the image

Input parameter type	Introduction	Example
byte[]	pixels,Image bytes, the field data represents the final data after the user processes the image	
int	startX,Start the position in the picture X direction, and adjust the picture display position	0
int	startY,Start the position in the picture Y direction, and adjust the picture display position	0
int	width,The width of the image	Global brush, take the width of the screen that the width of

		the screen is equal to the width of the picture, local brush take the width of the picture
int	height,The image of the height	Global brush, take the height of the screen that the height of the screen is equal to the height of the picture, local brush take the height of the picture
int	waveform mode	Color screen 0, other 0,1,2

Return value type	**Returned value specification**	**Examples**
int	0 Success, other reference status code	

3.17 Display

Function: public int display(DisplayParams displayParams)

Instruction: Display image

Input parameter type	**Introduction**	**Example**
DisplayParams	displayParams, Image display parameters	DisplayParams displayParams = new DisplayParams.Builder(startX, startY, width, height, waveformMode).build();

3.18 Display

Function: public int display(int startX, int startY, int imgW, int imgH, int waveform)

Instruction: Display image

Input parameter type	**Introduction**	**Example**
int	startX,Start the position in the picture X direction, and adjust the picture display position	0
int	startY,Start the position in the picture Y direction, and adjust the picture display position	0
int	width,The width of the image	Global brush, take the width of the screen that the width of the screen is equal to the width of the picture, local brush take the width of the picture
int	height,The image of the height	Global brush, take the height of the screen that the height of the screen is equal to the

		height of the picture, local brush take the height of the picture
int	Waveform mode	Color screen 0, other 0,1,2

Return value type	Returned value specification	Examples
int	0 Success, other reference status code	

3.19 Clear Screen

Function: public int clearDisplay(int startX, int startY, int imgW, int imgH, int waveform)

Instruction: Clear the screen, as shown in white

Input parameter type	Introduction	Example
int	startX,Start the position in the picture X direction, and adjust the picture display position	0
int	startY,Start the position in the picture Y direction, and adjust the picture display position	0
int	width,The width of the image	Global brush, take the width of the screen that the width of the screen is equal to the width of the picture, local brush take the width of the picture
int	height,The image of the height	Global brush, take the height of the screen that the height of the screen is equal to the height of the picture, local brush take the height of the picture
int	Waveform mode	Color screen 0, other 0,1,2

Return value type	Returned value specification	Examples
int	0 Success, other reference status code	

3.20 Firmware OTA upgrade

Function: public int upgradeFirmware(File file)

Instruction: Upgrade the T1000 firmware

Input parameter type	Introduction	Example
File	file, firmware file, the file name suffix is the bin	

Return value type	Returned value specification	Examples
int	0 Success, other reference status code	

3.21 Waveform OTA upgrade

Function: public int upgradeWbf(File wbfFile)

Instruction: Upgrade the T1000 waveform file

Input parameter type	Introduction	Example
File	file, firmware file, the file name suffix is the wbf	

Return value type	Returned value specification	Examples
int	0 Success, other reference status code	int

3.22 IT8951 Software Reset

Function: public int it8951SoftwareReset()

Instruction: Only support for IT8951 device soft reset

Return value type	Returned value specification	Examples
int	0 Success, other reference status code	int

4 Status Code

Description: The actual status code returned shall prevail. The following is the common status code

Status Code	Instruction
0	Success
-1	Communication timeout
-2	Failed to send data (write failure)
-3	Failure to receive data (read failure)
-4	Data validation failed
-5	The device is not connected
-6	The device is connected

-7	Data returned to empty
-8	Operation cancelled
-9	This feature is not supported
-10	Uninitialized
-11	The device was not found
-12	Device driver is not found
-14	No permission
-15	Permission rejected
-16	Open the failure
-17	Display out of bounds
-18	The file does not exist
-19	The file format error
-20	Display timeout
-21	Display exception
-99	Communication failure

5 Waveform Mode

Different EPD have different WF modes, colors EPD is only 0 mode, black and white EPD has 0,1,2 three modes, the following is the monochrome EPD WF mode introduction:

Mode	Value	Instruction	Supported temperature range
INIT (Global update WF)	0	**Initialize** It can be used to completely remove any images, such as lost records due to the restart.	0~50°
DU(Local update WF)	1	**Direct update** Non-flashing waveform that can be used to update. It can update any changed graytone pixel to black or white only.	-15~65°
GC16(Global update WF)	2	**Grayscale Clear, 16 Levels** A “flashy” waveform used for 16 level grayscale images. All the pixels are updated or cleared.	0~50°